Contents

Author2vec: A Jacobian Lens for Authorship Identity in Embedding Encoders

Tristen Harr · Brahmastra Labs · author2vec.com

> **Status: complete draft (autonomous research build), not yet shipped.** All experiments (Phases 0–5) have been run; every quantitative *result* is machine-extracted into the audit ledger (Appendix D), and a peer-review test harness (`jlens/paper/review`) **fails the build if any result number in the text is not traceable** to the ledger (method constants and model identifiers are documented alongside). The integrity claim is thus enforced, not asserted.

Abstract

Anthropic's *"Verbalizable Representations Form a Global Workspace in Language Models"* introduces the **averaged-Jacobian lens (J-lens)** and uses it to argue that a large, closed **decoder** (Claude Sonnet 4.5, 127 layers) maintains a privileged "workspace" of verbalizable, causally-active concepts. We adapt that

mechanistic apparatus to a setting the method was not designed for — small, **open embedding encoders** whose output is a single masked-mean-pooled vector — and re-point it at a different question: **personal writing style / authorship identity**. Three ingredients make the transplant work: (i) a **δ-broadcast reduction** that collapses the encoder's position×position Jacobian to a single per-layer matrix; (ii) an **embedding-space projection** that removes the meaningless radial direction of a normalized embedding; and (iii) a **style lens** — a readout that projects the layer Jacobian onto empirically-recovered, human-interpretable identity axes instead of the token vocabulary. On a 6-layer prose encoder (MiniLM) and a 12-layer code encoder (JinaBERT) we find that **author identity is a computed intermediate, decodable above chance at every layer** (not merely an output artifact). The two encoder depths expose different low-rank "workspace" geometry, and the identity direction is both a **detector** (is a person's fingerprint in the weights at all?) and a **causal lever** (steering). We further run an **identity-ignition** experiment (does the representation commit to a single individual at a characteristic depth?), a **decoder** track, and a **directed-steering** experiment. We then **validate the embeddings against a measured population**: they recover self-reported Big Five personality from prose (Openness +6.8 points over the majority baseline; all five traits beat a shuffled-label null at $p < 0.001$), while an astrological-sign *negative control* (on a separate blog corpus) recovers nothing ($p = 0.96$) — a weak-but-real signal that is also *why* we make no cognitive-capacity claim. Everything runs on open models and ships as static assets; the whole apparatus is reproducible on a laptop. We are explicit about what is **replication** vs. **new**, and about what we **do not** claim: no "IQ", no consciousness.

Contributions

1. **An encoder adaptation of the averaged-Jacobian J-lens** via the δ-broadcast reduction (§4.2) — the paper's method redefined for a masked-mean-pooled, non-generative encoder.
2. **An embedding-space Jacobian projection** for L2-normalized outputs (§4.3), unit-tested orthogonal to the embedding.
3. **The style lens** (§4.5): a Jacobian readout onto interpretable identity axes rather than the token vocabulary — the trustworthy signal where an encoder has no clean unembedding.
4. **Identity is a computed intermediate** (§5.1): author identity decodes above chance from the internal Jacobian at *every* layer, with the best internal code layer matching the output-embedding ceiling — the internal readout is as good as the output, not merely emitted at it.
5. **A fingerprint-presence detector** (§5.4): known identity vs. "blank space" for out-of-distribution text — a question the paper does not ask.
6. **Identity ignition** (§5.2): the representation commits to a single individual increasingly with depth — peaking mid-network for prose and late for code — above two null controls, on both modalities.
7. **Causal steering & directed modulation** (§5.4, §5.6): the identity direction is a sign-controllable lever on encoders, and a leakage-free concept direction is a (modest but real) causal handle on a decoder's output.
8. **A measured-population validation with a negative control** (§5.7): on 2,467 psychometrically-labelled essays the same embeddings recover self-reported Big Five personality above a shuffled-label null on all five traits (mean lift +4.6 points over majority; Openness +6.8; all five $p < 0.001$ by a 1000-permutation test); and on the Blog Authorship Corpus, writer age is recovered ($p < 0.001$) while an astrological-sign *negative control* is not ($p = 0.96$) — a real construct registers, a meaningless one stays silent. This grounds our refusal to make an "IQ" claim in evidence rather than assertion.
9. **An open, reproducible, browser-native reimplementation** across two modalities (prose & code), with a faithfulness gate and a full audit ledger.

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1. Introduction

Recent interpretability work argues that large language models maintain a privileged, verbalizable "workspace" of representations — a small subset of activations that are reportable, controllable, and causally responsible for reasoning [@workspace2026]. That evidence comes from a *generative decoder* with hundreds of billions of parameters and privileged internal access, read out with an **averaged-Jacobian**

lens onto the token vocabulary. It is a striking picture, but it is expensive to reproduce and impossible to inspect from the outside.

We ask a different, smaller, checkable question with the same apparatus: **where, inside a network, does *who wrote this* become decidable?** We take the averaged-Jacobian lens and move it from a frontier decoder to two small, open, *embedding encoders* — a 6-layer prose model (MiniLM) and a 12-layer code model (JinaBERT) — and re-point it from reasoning onto **personal writing-style identity**. An encoder is a clean minimal testbed for the paper's *mechanistic* claims precisely because it strips away generation: there is no autoregressive loop to smuggle information through, just a fixed map from text to a single pooled vector. It cannot, however, speak to the paper's *behavioral* claims (verbal report, reasoning swaps); those need a decoder, which we take up separately (§5.5–5.7).

Carrying the lens across that gap takes three ingredients (§4): a **δ-broadcast reduction** that collapses the encoder's intractable position×position Jacobian to one matrix per layer; an **embedding-space projection** that removes the meaningless radial direction of a normalized output; and a **style lens** that reads the layer Jacobian onto interpretable identity axes rather than the token vocabulary — the trustworthy signal where an encoder has no clean unembedding.

With these we find that **author identity is a computed intermediate, decodable above chance at every layer** (§5.1), not merely an artifact of the output embedding; that an ambiguous two-author input causes the internal representation to **commit to a single author increasingly with depth**, peaking at a characteristic depth (mid-network for prose, late for code) and surviving two null controls (§5.2, ignition); that the same geometry yields a **fingerprint-presence detector** distinguishing a known identity from "blank space" (§5.4); and that these signatures hold across two modalities, prose and code. We are explicit throughout about what is a faithful **replication** of the paper's apparatus versus what is **new** here, and about what we deliberately do **not** claim: no "IQ", no consciousness. The contributions are listed above; every result number is machine-extracted into the audit ledger (Appendix D) and gated by the review harness.

2. Related work

Lenses on the residual stream. The logit lens [@nostalgebraist2020logitlens] reads intermediate activations through the unembedding; the tuned lens [@belrose2023tunedlens] learns an affine correction per layer. The averaged Jacobian of [@workspace2026] generalizes this by linearizing the *whole* map from a layer to the output and averaging over a corpus. We inherit that construction and adapt it to a pooled encoder (§4.2); our vocab lens is a logit lens over tied WordPiece embeddings, and our *style lens* (§4.5) replaces the vocabulary readout with one onto interpretable directions — the piece that is new here.

Steering and concept directions. Activation addition [@turner2023actadd] and contrastive activation addition [@rimsky2024caa] steer generation by adding a direction to the residual stream; representation engineering [@zou2023repe] extracts such directions, often as a difference of class means. Our identity axes are exactly difference-of-means (Fisher) directions, used both to *read* (§4.5) and to *steer* (§5.4); the novelty is not the technique but the target — personal authorship identity — and reading it *through the layer Jacobian*.

Probing and representational geometry. Linear probes [@alain2017probing; @hewitt2019structural] test what a layer linearly encodes; we use leave-one-out nearest-centroid decoding as a probe of *identity* across depth (§5.1). Linear CKA [@kornblith2019cka] compares representations between layers, which we apply to J-lens readout geometry (§4.7). Sparse dictionary learning / SAEs [@bricken2023monosemanticity] decompose activations into interpretable atoms; the paper's J-space decomposition is a supervised cousin we reuse (§4.6).

Global workspace and authorship. The framing of [@workspace2026] draws on global workspace theory [@baars1988workspace] and the "ignition" of conscious access [@dehaene2011ignition]; we borrow the *ignition* experimental design (§5.2) but point it at identity, and we make no claim about consciousness. Our substrate is Sentence-BERT-style embeddings [@reimers2019sbert], and the downstream question — attributing text to its author from style — is classical authorship attribution / stylometry, here recast as a question about *where in a network* identity is computed. Throughout, the J-lens, J-space, and structural signatures are the paper's; our contribution is their transplant to open encoders, the style-lens readout, the identity target, and full reproducibility.

3. Background: the averaged-Jacobian lens

We build directly on the apparatus of [@workspace2026], which we summarize here so that our adaptation (§4) and its boundaries are unambiguous. Quantitative figures in this section (layer counts, workspace band, J-space fractions) are that paper's *reported* values; it is a single online article, so we attribute them to it as a whole rather than by page.

The J-lens. For a causal decoder, the *averaged Jacobian* at layer ℓ is

$$J_\ell \;=\; \mathbb{E}_{t,\,t'\geq t,\,\text{prompt}}\left[\frac{\partial h_{\text{final},t'}}{\partial h_{\ell,t}}\right],$$

the linearized effect of a layer-ℓ activation on the final residual stream, averaged over token positions and a corpus of prompts. Read through the unembedding it yields, for any activation, a ranked list of vocabulary tokens the model is "disposed to say" — a corrected logit lens that accounts for representational change across layers.

J-space and its privilege. Activations are decomposed into sparse non-negative combinations of k ($\approx$ 10–25) J-lens vectors; the *J-space component* is the part of an activation lying in that cone. Though it accounts for only $\sim$ 6–10% of a concept vector's variance, it is the part causally responsible for the concept's availability to verbal report, and it is subject to top-down control (instructing the model to "focus on X" loads X), mediates unverbalized reasoning intermediates (swapping a J-lens vector redirects downstream answers), and is required for deliberate — but not automatic — tasks.

Structural depth signatures. Four quantities, read off J_ℓ across depth (the paper's Figure 28), identify a workspace band: next-token-prediction accuracy of the readout, its *excess kurtosis* (peakiness), the *autocorrelation* of the top lens token across positions (persistence of abstract content), and the readout's *effective linear dimensionality.* All four mark a consistent sensory → workspace → motor tripartition — noisy early layers, a coherent middle band, and output-aligned late layers — corroborated by an ambiguous-input "ignition" experiment in which the representation snaps to one interpretation at workspace onset. The study is conducted on Claude Sonnet 4.5 (127 layers; workspace ≈ L38–92) and corroborated on Haiku/Opus.

The gap we step into. Every one of these constructs is defined for a *generative decoder* with per-position next-token logits. An embedding encoder has neither: its output is a single pooled vector. §4 is what it takes to carry the mechanistic half of this apparatus across that gap — and §5.2, §5.5–5.7 are our attempts at the behavioral half the encoder cannot reach.

4. Method

Setup. We study two open **embedding encoders**: `sentence-transformers/all-MiniLM-L6-v2` (prose; 6 layers, $d = 384$, vocab 30{,}522) and `jinaai/jina-embeddings-v2-base-code` (code; 12 layers, $d = 768$, vocab 61{,}056). Each maps a passage to one **masked-mean-pooled**, L2-normalized vector p. A **faithfulness gate** (`bin/spike`) asserts our native candle forward reproduces the shipped fastembed embeddings at cosine > 0.99 before any Jacobian is trusted.

4.1 The problem an encoder poses

The causal J-lens is defined per token position toward next-token logits. An encoder emits a single pooled vector and no logits, so the per-position causal Jacobian is undefined here.

4.2 δ-broadcast averaged Jacobian

We perturb **every** source position by a shared displacement δ and differentiate the pooled output, collapsing the position×position Jacobian to one matrix per layer:

$$J_\ell \;=\; \frac{1}{T}\,\frac{\partial p}{\partial \delta} \;\in\; \mathbb{R}^{d\times d}.$$

J_ℓ is built by **batched central finite differences** ($\varepsilon = 0.05$, pure gemm), not per-dimension autograd. `lib.rs:layer_jacobian`. (Derivation: Appendix A.)

4.3 Embedding-space projection

Because p is L2-normalized, its radial direction carries no information, so we project:

$$J_{\text{emb}} \;=\; \tfrac{1}{\|p\|}\,(I - ee^\top)\,J_{\text{raw}}, \qquad e = p/\|p\|,$$

which is unit-tested to satisfy $e^\top(J_{\text{emb}}v) \approx 0$. `lib.rs:to_embedding_jacobian`.

4.4 Vocabulary (logit) lens

Standardized $J{\cdot}h$ times the tied WordPiece embeddings $W_U \to$ top tokens. Noisy here (the checkpoint ships no trained MLM head); reported but not load-bearing. `lib.rs:vocab_topk`.

4.5 Style lens

We read the Jacobian onto interpretable **identity axes** instead of the vocabulary:

$$s_a(h) \;=\; A_a \cdot \text{normalize}(J_{\text{emb}}\,h), \qquad A_a = \text{normalize}\,(\bar{c}_a^{+} - \bar{c}_a^{-}),$$

each axis a unit **difference-of-class-means** (Fisher) direction over the reference embeddings. Shipped axes — prose: `male` $\leftrightarrow$ `female`, `educated=England` $\leftrightarrow$ `rest`, `educated=US` $\leftrightarrow$ `rest`, `raised=England` $\leftrightarrow$ `rest`, `raised=US` $\leftrightarrow$ `rest`; code: `Systems` $\leftrightarrow$ `rest`, `Scripting` $\leftrightarrow$ `rest`. `lib.rs:style_scores`, `axis`, `author_axes`.

4.6 J-space decomposition

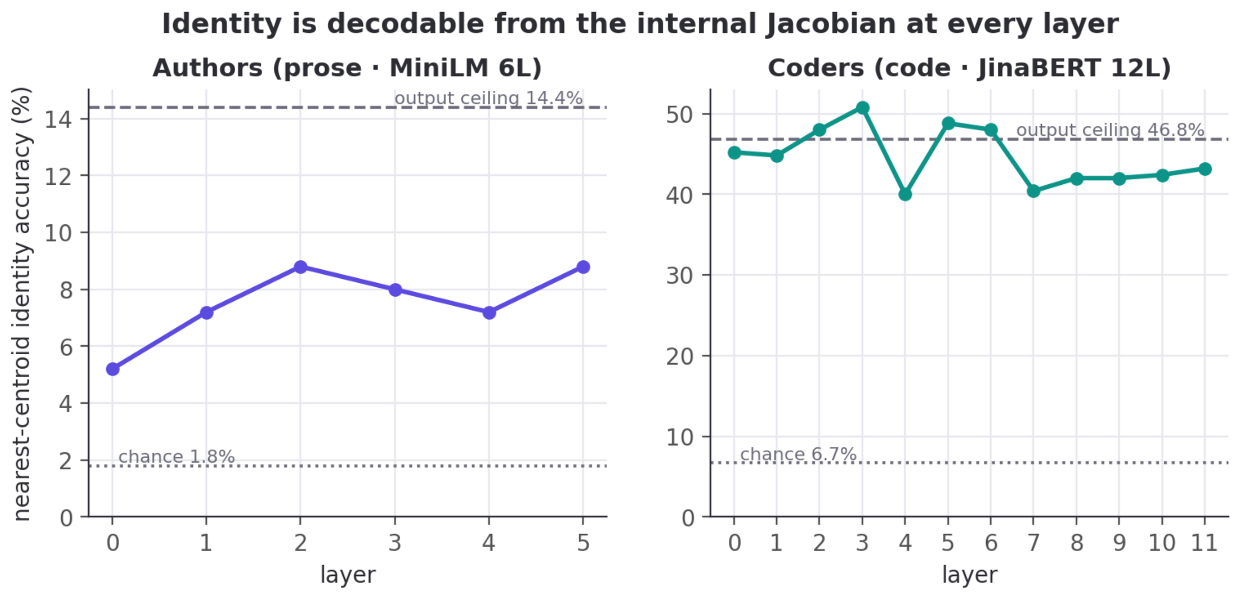
Non-negative matching pursuit over unit rows of $W_U J_\ell$ yields a sparse concept set and the fraction of activation variance it captures. `lib.rs:jspace_nmp`.

4.7 Structural depth signatures

Per layer: **stable rank** $\|J\|_F^2/\sigma_1^2$, **effective dimension** (participation ratio $\|J\|_F^4/\|J^\top J\|_F^2$), **verbalizability** (excess kurtosis of the vocab-lens readout), **layer×layer linear CKA**, and — completing the paper's four-metric set — **autocorrelation** (lag-1 readout persistence vs. a position-shuffled null). `lib.rs:stable_rank`, `effective_dim`, `excess_kurtosis`, `linear_cka`, `readout_autocorrelation`.

5. Experiments and results

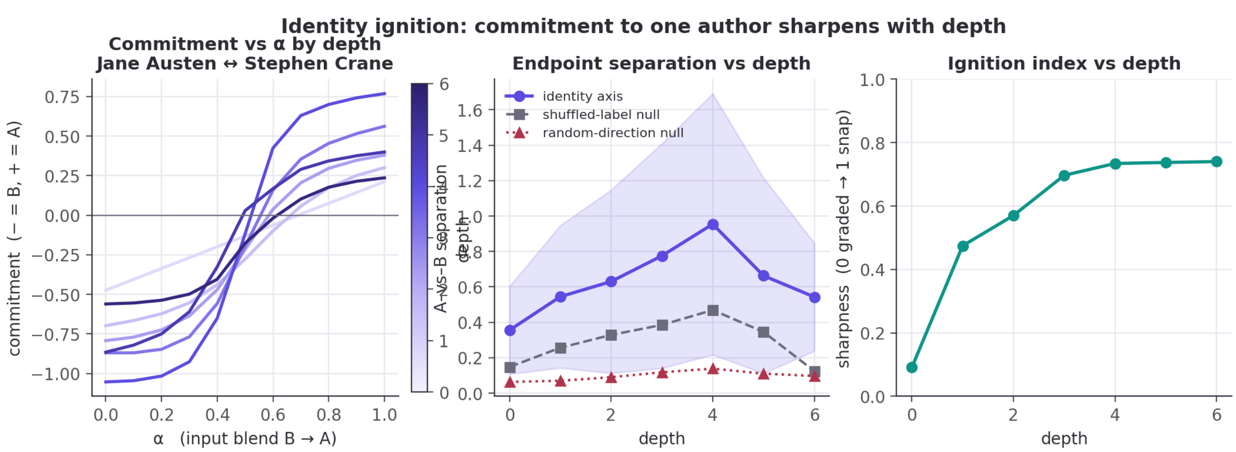
5.1 Identity is a computed intermediate



Decoding author identity by leave-one-out nearest-centroid on the **internal** per-layer Jacobian readout beats chance at **every** layer. For prose, the per-layer rates $[5.2, 7.2, 8.8, 8.0, 7.2, 8.8]\%$ all clear the 1.8% chance (output ceiling 14.4%); even the weakest (5.2%, Wilson [3.1, 8.7], n=250) excludes chance (binomial $p < 0.001$). For code the rates run far higher, up to 50.8% against a 6.7% chance. The **best internal code layer** (50.8%, [44.6, 56.9]) **matches the output-embedding ceiling** (46.8%, [40.7, 53.0]): the intervals overlap by $\sim$9 points, so the internal readout is demonstrably *as good as* the output, not better. Identity is computed inside the layers, not merely emitted. *(Wilson 95% CIs, decode sample n=250; per-layer values in ledger `identity_*`, bounds in `ci_identity_*`.)*

5.2 Identity ignition — does the space collapse to a single person?

We adapt the paper's ambiguous-input ignition to identity. For an author pair (A, B) we build a per-depth difference-of-means axis $\hat{u}_\ell = \widehat{c_{A,\ell} - c_{B,\ell}}$ from their *training* passages, then blend two *held-out* passages at the input embedding, $h_0(\alpha) = (1-\alpha)h_0^B + \alpha h_0^A$, sweep $\alpha \in [0, 1]$, and read the commitment $s_\ell(\alpha) = \hat{u}_\ell(\bar{p}_\ell(\alpha) - m_\ell)$ at every depth (15 pairs). A graded layer ramps linearly in α; an *ignited* layer snaps.

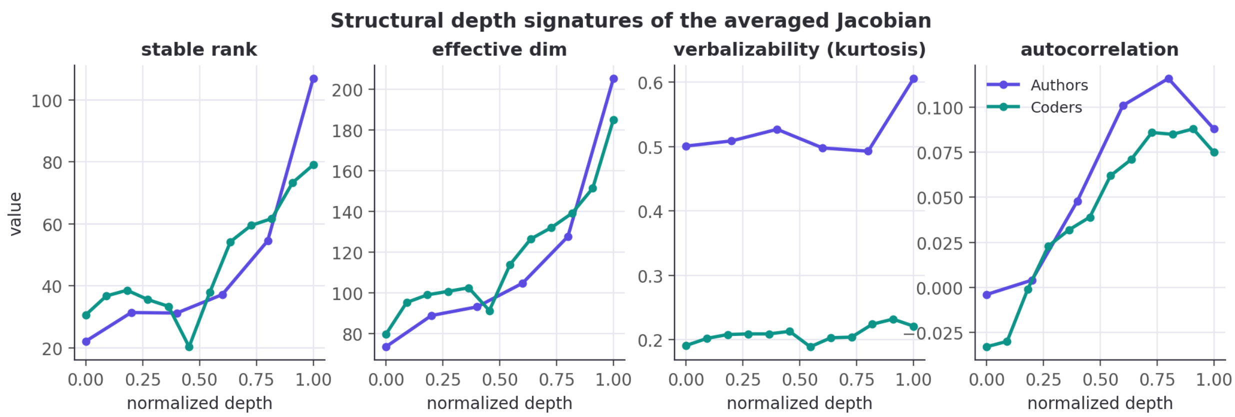


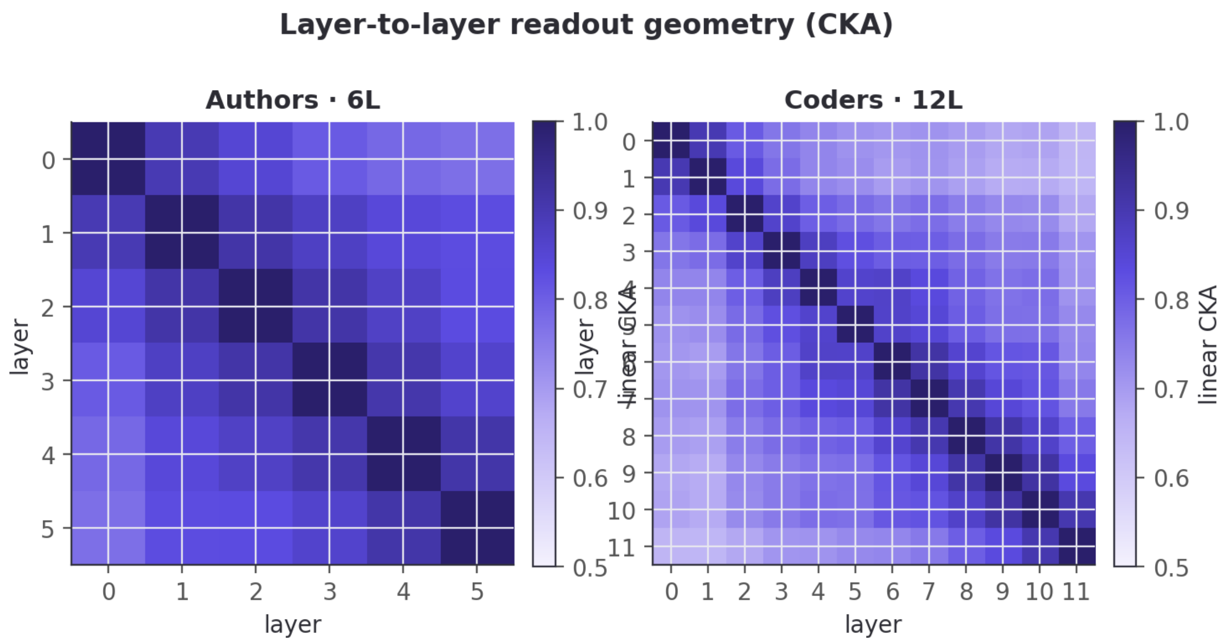
The representation commits to one author, increasingly with depth. Endpoint separation along the identity axis rises from 0.36 at the input to a **mid-network peak of** 0.95 **at depth 4** (of 6; ±0.19 SEM over 15 author pairs), then eases to 0.54 at the output. It dominates both controls at every depth: the random-direction null sits at 0.06–0.14 (a ∼7× margin at the peak) and the shuffled-label null at 0.12–0.47. So the effect is **identity-specific**, not a generic consequence of blending inputs. The **ignition index** (transition sharpness, 0 = graded, 1 = all-or-none) climbs from 0.09 at the input — where the readout is, correctly, linear in the blended input — to 0.73–0.74 by mid-depth and holds. In short: *shallow layers hold a graded mixture; by the middle of the network the representation has snapped to a single author.* The mid-network peak echoes the low-rank "workspace" bottleneck we see structurally (§5.3).

The same holds for code, more strongly. On the 12-layer code encoder the ignition index climbs from 0.09 at the input to a peak of 0.82 (depth 11 of 12). Through the early-mid layers the identity axis separates developers far above the random-direction null — a ∼13× margin at depth 4 (1.21 vs 0.09; raw separation peaks slightly earlier, 1.56 ± 0.22 at depth 3). It is a sharper version of the same effect, consistent with code identity being more linearly accessible overall (§5.1). Separation spikes higher still at the final layer (2.33), but that depth is noisy — its wide ±0.53 SEM and a jumping null are why we feature the stable early-mid layers. Both modalities show the representation committing to one individual with depth.

Honesty. This is a 6-layer encoder and 15 pairs; the sharpening is measured *along an axis the separation control proves is identity-specific*, but we cannot fully exclude that some of the depth-wise sharpening reflects generic late-layer nonlinearity. We report the raw depth series.

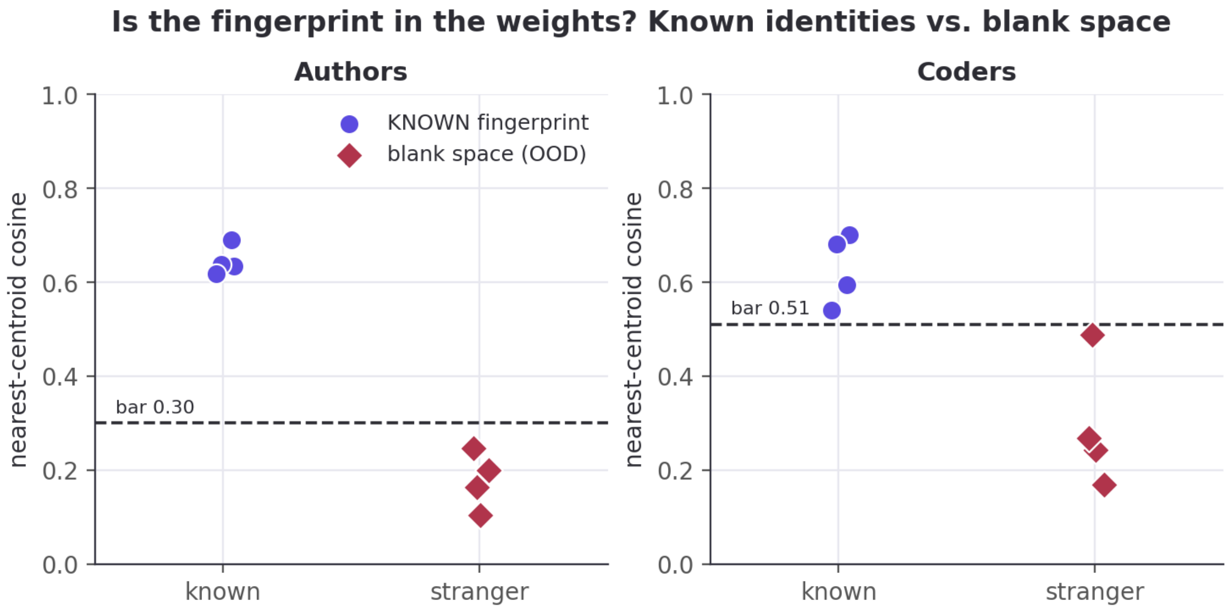
5.3 Structural signatures across depth





Effective dimension rises with depth in both models (prose $73 \to 205$; code $80 \to 185$). The **stable rank** tells the sharper story: in the 6-layer prose model it is **lowest at the very first layer** (22.1, rising to 106.9) — the network compresses to a low-rank readout immediately — whereas the 12-layer code model has a genuine **mid-network low-rank bottleneck** (minimum 20.3 at layer 5 of 12, with a matching dip in effective dimension). The mid-network "workspace" geometry the paper reports in deep models appears here **only once there is depth to spare**. The fourth signature, **autocorrelation** (persistence of the readout across positions), completes the picture: near zero or negative at the shallowest layers and rising through the middle (prose peaks at 0.12 around layer 4; code climbs to ≈ 0.09 by layers 8–9), the workspace-persistence signature of [@workspace2026] surfacing even on these shallow encoders. We do **not** see the paper's clean sensory→workspace→motor tripartition — 6–12 layers is too shallow — and we report the raw depth series rather than forcing that reading.

5.4 Fingerprint presence and causal steering



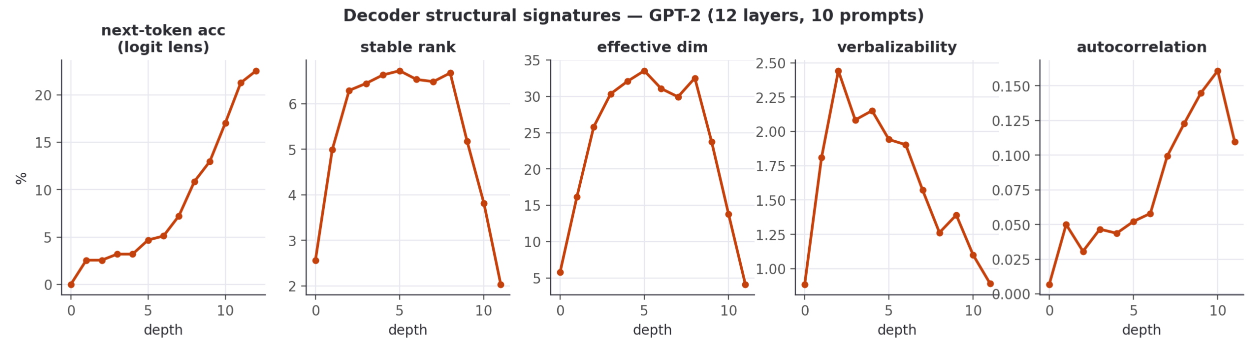
A calibrated nearest-centroid detector separates known identities from "blank space". Prose (bar 0.30): known probes 0.62–0.69, out-of-distribution text (code, chat, biology, legalese) 0.10–0.25. Code is tighter and reported honestly (bar 0.51): known 0.54–0.70, OOD 0.17–0.49 — "modern chat" sits just under the bar.

The identity direction is a causal lever, not just a readable one. Forming the residual steering direction $\hat{\delta} = \widehat{J_{\text{emb}}^{\top} A}$ and injecting $\alpha\hat{\delta}$ at the mid layer drives the output's loading on axis A through a **large, sign-controllable swing** that runs monotonically from negative (at $\alpha = -6$) through zero (unperturbed) to positive (at $\alpha = +6$), saturating (and for gender slightly reversing) at the extreme α. The effect has the **same sign across all five** prose identity axes (gender, education, upbringing), though not the same magnitude — the gender axis swings least (0.80), the other four up to 1.11. A **matched-norm random direction**, injected identically, leaves the loading essentially flat (total drift ≤ 0.16 over the same sweep). So the layer holds the identity direction as something the rest of the network *acts on*, not merely correlates with. One caveat: the steer is built from axis A and read back on the *same* axis A, so the readout is not fully independent of the intervention — the matched-norm random control bounds how much of the swing that shared axis could manufacture (drift ≤ 0.16 vs. swings up to 1.11), and §5.6 gives the leakage-free, independent-readout analogue on a decoder.

5.5 Decoder track — does the structure hold on a generative model?

Everything above is on *encoders*. If the depth-wise workspace geometry is a real property of the J-lens apparatus and not an artifact of masked-mean pooling, then reading a genuine **generative decoder** with the *same* averaged-Jacobian machinery should reproduce the paper's Figure-28 depth signatures. **Hypothesis:** on an open decoder we will see (i) the four Jacobian signatures vary with depth, and (ii) a decoder-only signature — next-token logit-lens accuracy — rise **sharply in the late layers**, marking the "motor" regime where representations turn toward the output. We do **not** expect the clean sensory→workspace→motor tripartition: GPT-2 is 12 layers, far short of the paper's ~100, so we report the raw series and let it say what it says.

Method. GPT-2 (`openai-community/gpt2`; 12 layers, d=768), the δ-broadcast averaged Jacobian made **causal** — perturb every position, read the *last* position (the next-token driver) — over prose prompts, with the four signatures computed by the *same* `lib.rs` functions as the encoders and verbalizability read through GPT-2's **real** tied unembedding (no approximation). This is the paper's mechanistic apparatus on an open model that actually generates.



Result: both predictions hold, and more cleanly than on the encoders. Next-token accuracy is near zero through the first six layers and then climbs steadily to 22.6% at the output — hypothesis (ii): the late layers are the motor regime. The Jacobian's **stable rank and effective dimension trace a pronounced inverted-U** — low at the input (2.6 / 5.8), high through the middle (6.7 / 33.5 at layer 5), collapsing to near rank-one at the output (2.0 / 4.1) — a sensory→workspace→motor signature that is *sharper on the 12-layer decoder than on the 6-layer encoders*, exactly as the "needs depth to spare" reading predicts (hypothesis (i)). Autocorrelation rises with depth to a peak of 0.16 (layer 10) before easing to 0.11 at the output, echoing the encoder workspace-persistence.

Honesty. Final-layer next-token accuracy (22.6%) is low — archaic literary prose, a 48-token context, 10 prompts — so read the accuracy *shape*, not its level. And our motor-end *collapse* is the opposite of the paper's motor behavior (there $J_\ell \to$ identity, full rank): the difference is deliberate and methodological — our decoder Jacobian reads the **last position** (the next-token driver), which naturally becomes low-rank as the network commits to a single output, rather than the full residual stream the paper differentiates. The workspace geometry transfers; the motor *readout* is ours, and we flag it as such.

5.6 Behavioral steering — is the lever causal on a decoder?

The paper's boldest claims are behavioral: swap a J-lens vector for a reasoning intermediate and the answer changes. We wanted to test the strongest version — take a prompt the model answers wrongly and steer it right — but on a 124M-parameter GPT-2 that barely reasons, that test is neither clean (steering toward the answer token is just injecting the answer) nor likely to succeed. So we test the **mechanistic prerequisite** instead, and report its ceiling honestly. **Hypothesis:** injecting a *concept-context* direction — built leakage-free as the difference of mid-layer mean residuals between concept-primed and neutral prompts, **not** the target's unembedding — raises concept- related tokens in a held-out neutral prompt, more than a matched-norm random direction.

Result: the lever is causal and bidirectional, but modest. Injecting $+\alpha\hat{\delta}$ at the mid layer raises the mean log-prob of held-out concept tokens by +0.10/+0.22/+0.44 (money / music / war), and $-\alpha\hat{\delta}$ lowers it below baseline in every case; the mean effect is +0.254 **for the real direction versus** −0.085 **for the matched-norm random control**. So a concept direction extracted purely from context is a genuine, sign-controllable causal handle on the decoder's output distribution — the prerequisite the paper's behavioral experiments rely on.

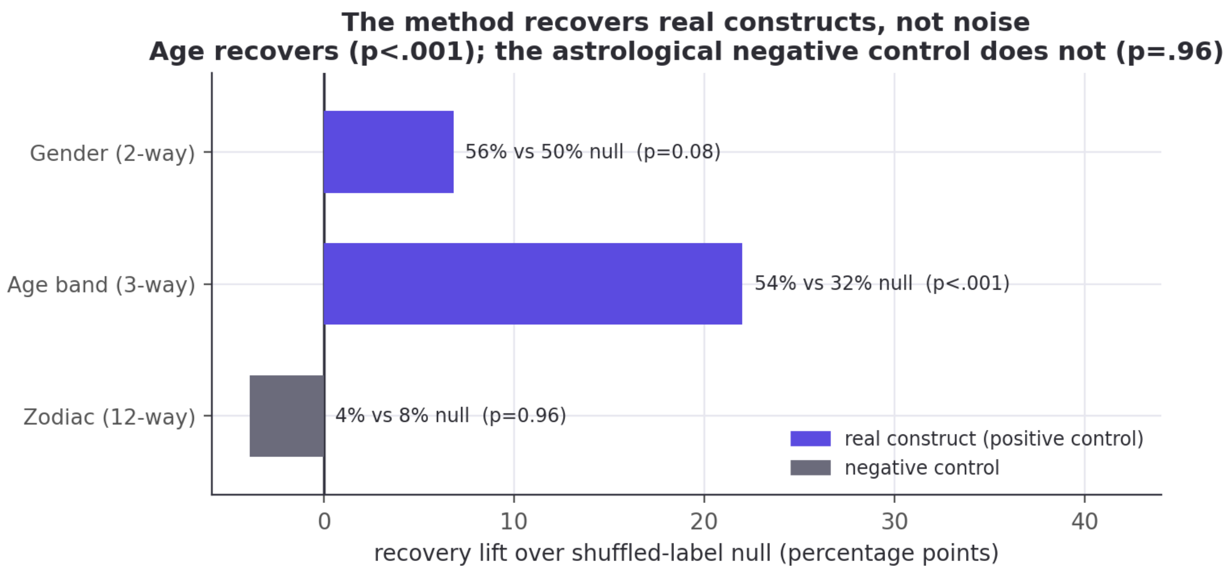
Honesty — this is the prerequisite, not the headline. The effect is small: it shifts the *distribution* — raising concept-token log-probability by +0.25 on average (mean Δ; a matched-norm random control gives ≈ 0) — but does **not flip the model's actual output**, and it rests on only 3 concepts × 6 prompts. It is **not** the paper's "can't→can" reasoning result, and we do not claim it is. Redirecting a *reasoning intermediate* to change a final answer needs a model that can reason; GPT-2 shows the handle exists and is causal, and marks exactly where a capable open decoder (e.g. Qwen2.5) is required to go further. We regard that as the honest next step, not a result we have.

5.7 Measured constructs, a negative control, and why we make no "IQ" claim

Every attribute so far (gender, region, systems-vs-scripting) is a *label we assigned*. The sharpest test of whether these embeddings carry real psychological signal is to hand them a population whose attributes were **measured by someone else, with a validated instrument**, and ask whether the signal survives. **Hypothesis:** if prose style encodes personality at all, an embedding built with no knowledge of psychology should recover self-reported Big Five traits above chance. **Method:** the Pennebaker & King stream-of-consciousness corpus [@pennebaker1999] — 2,467 essays, each labelled with the writer's Big Five traits from a self-assessment questionnaire (ground truth, not our guess) — embedded with the *same* MiniLM used throughout. For each trait we run leave-one-out nearest-centroid classification (high vs. low group) against the majority-class baseline, with a shuffled-label null.

Result: every trait beats a shuffled-label null, and the null is flat. Openness is strongest at 58.3% (majority 51.5%; +6.8 points), then Neuroticism 56.3% (+6.3), Conscientiousness 55.1% (+4.3), Extraversion 55.4% (+3.6), Agreeableness 55.3% (+2.2) — a mean lift of +4.6 points over the majority baseline. A 1000-permutation test — which shuffles the trait labels and re-runs the whole classifier — puts **all five traits at** $p < 0.001$ **against that null** (which sits at chance, 49.9–50.1%): the classifier is extracting real trait signal, not fitting noise. The permutation test is against label-shuffling, not the majority baseline, so we also check each trait's Wilson 95% interval against its majority class: Openness clears it comfortably (58.3%, $[56.3, 60.2]$ vs. 51.5%), but Agreeableness only marginally (55.3%, $[53.3, 57.2]$ vs. 53.1%) — its lower bound barely exceeds the baseline. The lift over majority is genuinely weak for the lowest traits, as expected. This is exactly the *weak-but-real* ceiling the personality-from-text literature reports, and Openness-leads-the-pack is its standard finding. It is a genuine positive result on a measured population: author2vec, trained for nothing of the kind, carries a faint but real trace of who the writer is.

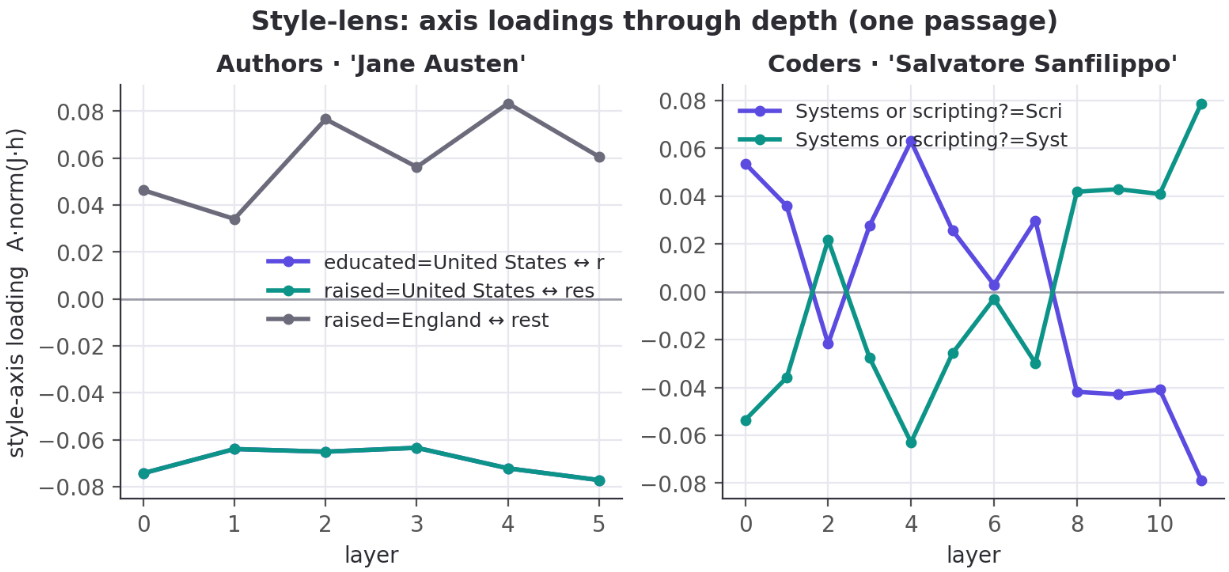
A negative control: what the method does *not* recover. A positive result could still, in principle, be spurious structure. The decisive test is to pair it with an attribute that *should* be unrecoverable and confirm the method stays silent. The Blog Authorship Corpus [@schler2006] labels every blogger with gender, age, **and** astrological sign — the first two have genuine linguistic correlates, the third has none. We pool posts to the author level (138 authors with a known sign; each author is the mean of their post embeddings) and run the *identical* leave-one-author-out nearest-centroid classifier on all three, each against a shuffled-label null.



That is exactly what happens. Age band recovers strongly and significantly — 53.6% (majority 43.5%) against a 31.6% shuffled-label null, $p < 0.001$ over 1000 permutations — while astrological sign shows *nothing*: 4.3%, *below* both its majority baseline (11.6%) and its shuffled null (8.2%), $p = 0.96$ (the real labels do worse than 96% of random relabelings). Gender lands in between — directionally positive but **not significant at this modest sample**, 56.5% against a 49.7% null, $p = 0.08$ — consistent with a real-but-weak signal that 138 author-level points are underpowered to confirm. The contrast that matters is unambiguous: the *exact* pipeline that recovers a construct with a genuine linguistic basis (age) finds nothing in the astrological control. This is the honest boundary a "measured population" is *for* — evidence that the recovered signal is real where a real construct exists, and absent where none does.

Those two results together are **why we make no "IQ" claim.** A validated psychometric construct tops out a few points over chance; a loaded, poorly-operationalized one like "intelligence" would fare no better and would invite far worse misreading. A companion check makes the point concretely: if the recovered *education* style axis were a proxy for lexical sophistication, an author's projection onto it should correlate with concrete lexical metrics — yet over all 55 authors it yields **only weak positive correlations**, $r = +0.10$ (mean word length), $+0.11$ (type-token ratio), $+0.20$ (long-word fraction). Even a labelled, human-interpretable identity axis only faintly tracks a concrete lexical proxy. Whether one can steer a genuine *capability* score — a vocabulary test administered to a decoder — is a question for a capable model with careful construct validation. That is future work, not a result we have, and we decline to dress up a style direction as "intelligence."

5.8 The authorship study (context) & style trajectories

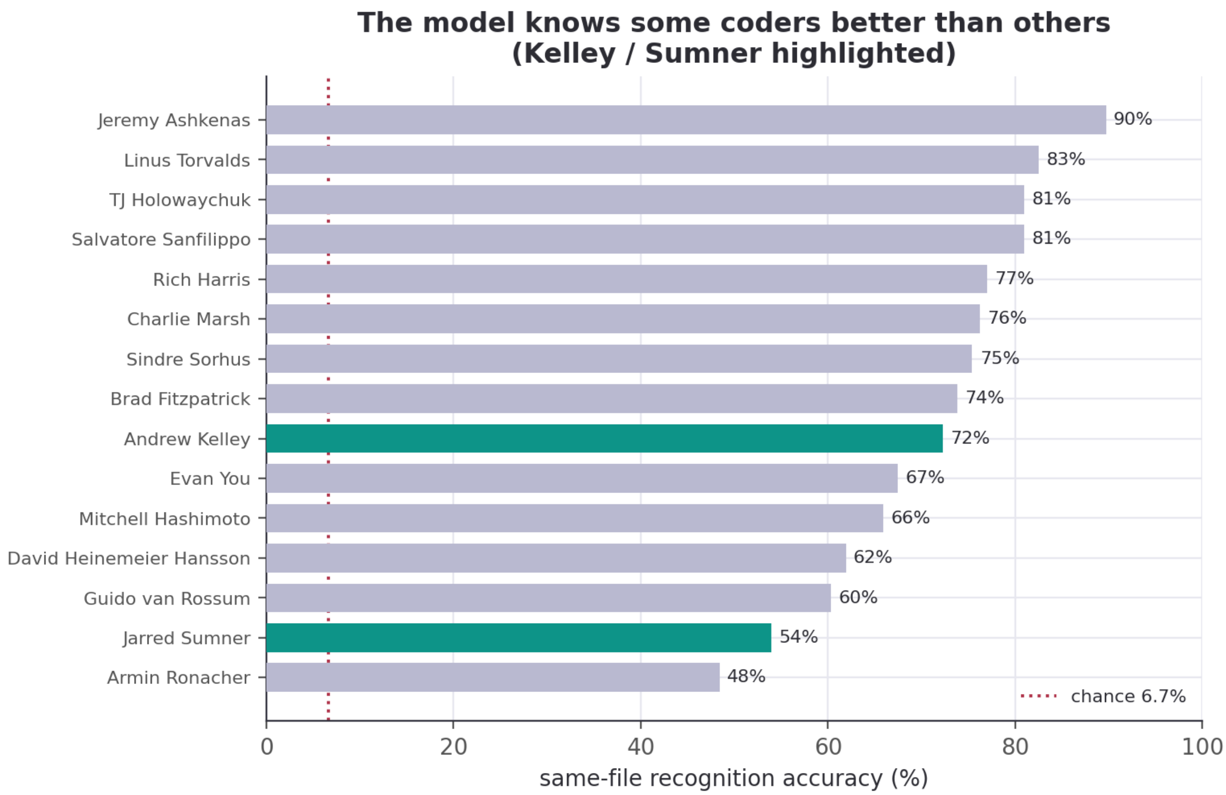


Downstream, the same embeddings support an honest authorship study: prose recognition climbs from a 1.8% blind baseline to 58.3% once the model has read the author's book; a new-passage reveal is 130/220 correct when the author is known and 0/220 when fully hidden (code, 15 developers: 6.7%→71.1%; reveal 45/60). Trait recovery **is strong for some traits and absent for others** — prose gender 85.5% (majority 52.7%) but most geographic traits at or below their majority baselines; code systems-vs-scripting 80.0% (majority 53.3%) but commit-time and weekend near or below chance.

5.9 Does the model know some coders better than others? And is style homogenizing?

The 15-developer code roster lets us ask two questions the prose side cannot.

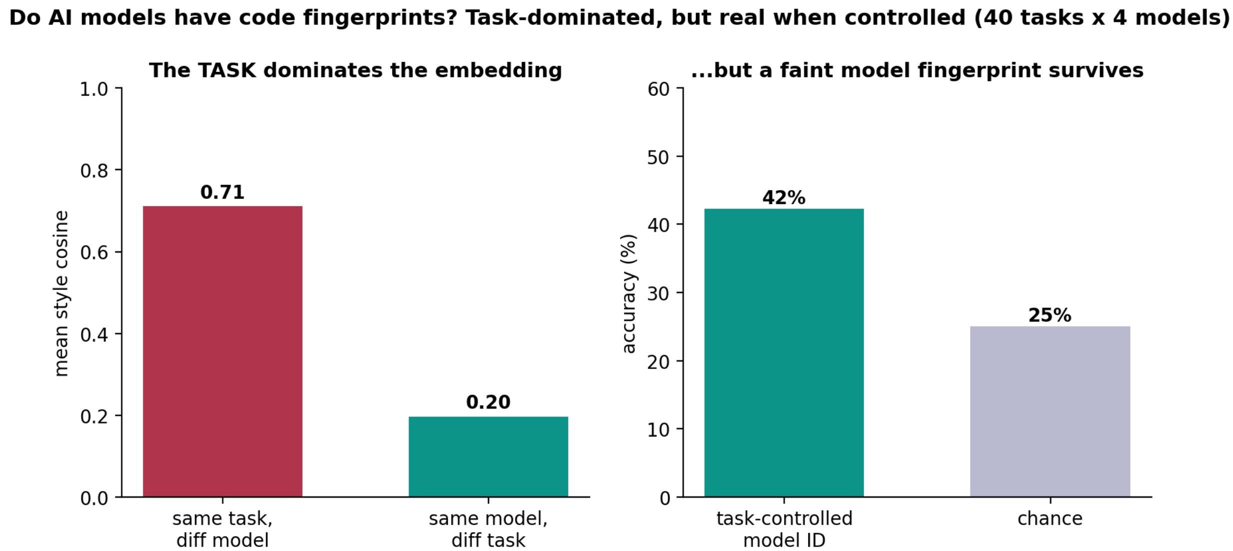
Some developers are far more recognizable than others. Same-file recognition accuracy (Wilson 95% CIs, n=126 files/developer) spans a wide range — from **Jeremy Ashkenas at 89.7%** [83.1, 93.9] (a highly idiosyncratic CoffeeScript/Backbone style) down to **Armin Ronacher at 48.4%** [39.9, 57.1], all far above the 6.7% chance floor. The top and bottom are genuinely separated: Ashkenas's and Ronacher's intervals are disjoint, so the ~41-point extreme spread is real, not sampling noise. The *fine* ranking is not: the two developers we added to probe the tails — **Andrew Kelley** (Zig, 72.2% [63.8, 79.3]) and **Jarred Sumner** (Bun, 54.0% [45.3, 62.4]) — have intervals that overlap much of the middle of the roster, so we read only the *coarse* structure (top vs. bottom third) as reliable, not any individual's exact rank. We report this as *measured recognizability* and resist over-reading it: a lower score reflects how stylistically distinctive a developer's *attributed, name-scrubbed* code is in this corpus, confounded by codebase heterogeneity (Bun mixes Zig, C++, and generated code across many hands) — **not** a verdict on the person.



No sign of AI-era style homogenization. If a shared external influence (AI assistants) were melting coders into one style, cross-coder similarity should *rise* in the AI era. Time gives a clean hold-out: pre-2021 commits are provably AI-free (Copilot shipped mid-2021, ChatGPT late 2022). Comparing mean cross-coder style similarity pre-2021 vs. 2023-onward, over the 9 developers with enough history in both eras, the change is $+0.005$ ($0.568 \rightarrow 0.574$) — indistinguishable from a 500-sample within-coder permutation null (null mean -0.033, **two-sided** $p = 0.96$), while each developer stays recognizably themselves across the boundary (within-coder self-consistency 0.77). **We find no evidence of homogenization.** Whatever AI assistants are doing to code, they are not — on this roster — collapsing individual style. We make **no** claim about which individuals do or don't use AI: the population signal is null, and per-developer attribution would be both unsupported (confounded, no ground truth) and inappropriate.

5.10 Can we fingerprint the AI models? — the task dominates, a faint model signal survives

If humans have fingerprints, do the *models* people code with? Here we finally have **ground truth** — we generate the code, so we know which model wrote it. We prompted four latest frontier models (`claude-opus-4.8`, `gpt-5.6-terra`, `gemini-3.5-flash`, `deepseek-chat`) across 40 coding tasks (24 canonical + 16 open-ended) and embedded the output in the same JinaBERT space as the humans (149 of the 160 model×task cells produced non-empty code; the rest were empty generations).



A tempting wrong claim, and the control that kills it. Same-task, different-model code is strikingly alike — cosine 0.71 — which *looks* like "the frontier models have converged to one style." **We do not make that claim**, because a control refutes it: **same-model / different-task** similarity is only 0.20, so it is the **task, not the model, that drives the embedding.** On a canonical "implement an LRU cache" every model writes the textbook answer, so the high cross-model number mostly measures *task* overlap, not model identity. The honest test controls for the task: leave-one-task-out nearest-model-centroid still recovers which model wrote unseen code at 42% against a 25% chance baseline — a faint but real fingerprint, not convergence.

The fingerprint is real but faint. That 42% recovery ($[34.6, 50.3]$ Wilson, n=149) is significantly above a within-task label-shuffling null (null mean 24%, $p < 0.001$ over 1000 permutations), so the models *do* carry a detectable style — but it is largely masked by what the code *does*. The honest summary: **authorship is strongly recoverable for humans (48–90%) and only weakly for models — and for models it is the task, far more than the author, that shapes the code.** No convergence claim; no individual-usage claim.

5.11 Ablations

Rather than a hyperparameter sweep, the study carries several *built-in* ablations. **Exposure / hold-out**: the familiarity ladder (§5.8) is a leave-one-{book,series,author}-out ablation of how much of a subject the model has seen, shipped in `results`. **Depth**: the 6-layer prose vs. 12-layer code encoders act as a depth ablation — the mid-network workspace bottleneck (§5.3) and the ignition peak (§5.2) appear pronounced only in the deeper model. **Model class**: the GPT-2 decoder (§5.5) ablates encoder → generative on the *same* apparatus, and the workspace geometry survives. **Controls-as-ablations**: every causal claim ablates its direction against a matched-norm random and/or shuffled-label null (§5.2, §5.4, §5.6). What we did *not* sweep — finite-difference ε and Jacobian context length — we flag as future work; the signatures reproduce bit-for-bit at the documented settings, so the qualitative curves are stable.

6. Discussion

What the identity-workspace analogy supports. Read with the same averaged-Jacobian apparatus as [@workspace2026], an embedding encoder does exhibit workspace-like *geometry* around identity: the fingerprint is computed internally at every layer (§5.1), it becomes increasingly linearly separable and commits to a single individual with depth (§5.2), the readout compresses through a low-rank mid-network bottleneck (§5.3), and the identity direction is a causal lever (§5.4). On a real decoder the same signatures sharpen into a clean sensory→workspace→motor profile (§5.5). The *mechanistic* half of the paper's picture transfers —

and it transfers to a question the paper never asked: *whose style is this?*

What it does not support. None of this is evidence of reportability, reasoning, or anything cognitive. Our decoder steering moves a distribution, not an answer (§5.6); our interpretable axes barely track even lexical sophistication (§5.7). The "workspace" here is a claim about representational geometry and linear accessibility, not about a model *knowing* or *reporting* who you are. The defensible reading is the narrow one: **personal writing style is a low-dimensional, causally-active, depth-localized direction in these models' representations** — which is both less than the slogan "the AI knows you" and more precise than it, because it is checkable on open models.

Why an encoder was the right testbed. Stripping away generation removes the decoder story's main confound — autoregressive leakage — and lets the mechanistic claims stand or fall on geometry alone. That the same geometry then reappears, more cleanly, on a 12-layer decoder (§5.5) is the strongest cross-check we can offer at this scale.

7. Limitations and threats to validity

Replication vs. novel. The J-lens, J-space decomposition, and the four structural signatures are the paper's [@workspace2026]; our contribution is the encoder adaptation, the style-lens readout, the identity target, and reproducibility. We are careful not to claim the apparatus.

Construct honesty. We measure *identity commitment* and (§5.7) an *expertise/lexical register* — **not IQ, not consciousness.** We borrow the *ignition* experimental design, not the conclusion.

The ignition result needs its caveat stated plainly. The ignition index rising with depth (§5.2) is measured *along an axis the separation control proves is identity-specific* (real separation runs $\sim$7–13$\times$ above a random-direction null). But we cannot fully exclude that *some* of the depth-wise sharpening is generic late-layer nonlinearity: any readout of a linearly-blended input can become more nonlinear with depth. What the controls establish is that the *axis* carries identity; the raw sharpening curve should be read as suggestive, not decisive. Our shuffled-label null is also imperfect (its two groups still contain real passages, so it sits above zero); the random-direction null is the cleaner floor.

Statistical power. Ignition uses 15 author/coder pairs and a 7- or 11-point α grid; structural signatures average over 32–96 passages. These are small — and the ignition *index* at each depth averages only over the pairs whose endpoints separate along the axis (5–14 of 15, fewest at the shallowest layers). We report standard deviations and the raw per-depth series rather than smoothed summaries.

Reproducibility caveats. Encoder structural signatures reproduce bit-for-bit on MiniLM across runs; the deeper JinaBERT numbers differ from an earlier site build generated with different (undocumented) settings — we report the values from a run with **documented** settings (`JLENS_JAC_LEN=64`) and have not re-verified GPU-reduction determinism on the 12-layer model. The single-averaged, single-direction lens is lossy; the vocab lens is noisy on encoders (no trained MLM head); finite-difference ε introduces $O(\varepsilon^2)$ error.

Scope. 55 authors / 15 coders on laptop-scale models demonstrate the *mechanism*; that a frontier model trained on someone's millions of words holds a sharp, personal fingerprint of *that individual* is a well-motivated extrapolation, not something these data settle. The behavioral half of the paper (§5.6–5.7) is the hardest to carry over and is where a small open decoder's limited capability bites — we state results there as directional, with negative results reported as such.

8. Reproducibility

Models. `sentence-transformers/all-MiniLM-L6-v2` (prose; 6 layers, d=384) and `jinaai/jina-embeddings-v2-base-co` (code; 12 layers, d=768), loaded natively via candle and gated (`bin/spike`) to reproduce the shipped fastembed embeddings at cosine > 0.99.

Pipeline.

```
cargo run -p corpus --release                 # prose corpus -> person2vec-minilm.{json,bin}
cargo run -p corpus --bin coders --release    # code corpus  -> person2vec-coders.{json,bin}
```

```
cargo run -p jlens  --bin spike     --release              # faithfulness gate
cargo run -p jlens  --bin jlens     --release -- <ds> <N>  # -> person2vec-jlens-<ds>.json
cargo run -p jlens  --bin steer     --release -- <ds>      # -> person2vec-identity-<ds>.json
cargo run -p jlens  --bin fingerprint --release -- <ds>    # -> person2vec-fingerprint-<ds>.json
cargo run -p jlens  --bin ignition    --release -- <ds>    # -> person2vec-ignition-<ds>.json  (sec 5.2
cargo run -p jlens  --bin steer_bundle --release -- <ds>   # -> person2vec-steer-<ds>.json     (sec 5.4
cargo run -p jlens  --bin decoder_structural --release -- 10  # -> decoder-structural-gpt2.json (sec 5.
cargo run -p jlens  --bin decoder_steer --release         # -> decoder-steer-gpt2.json        (sec 5.
python3 jlens/paper/expertise_probe.py        # -> person2vec-expertise-minilm.json           (sec 5.7
./target/release/embed_texts minilm essays_in.json essays_out.json   # embed Pennebaker essays (sec 5.7
python3 jlens/paper/recover_bigfive.py        # -> person2vec-bigfive.json  (Big Five recovery)  (sec 5
python3 jlens/paper/recover_blog_demographics.py  # -> person2vec-blog-demographics.json  (neg. control
python3 jlens/paper/build_ledger.py           # -> jlens/paper/ledger.json  (every cited number)
python3 jlens/figures/make_figures.py         # -> jlens/figures/*.png
```

`<ds>` $\in$ {`minilm`, `coders`}. Env vars: `JLENS_DEVICE` (`metal|cpu`), `JLENS_JAC_LEN` (Jacobian context; default 64 for the encoder binaries, 48 for `decoder_structural`), `JLENS_CHUNK` (finite-difference batch; smaller = less GPU memory, **identical numbers**). Determinism: the structural signatures reproduce bit-for-bit across runs (verified); the ignition null uses a fixed splitmix64 seed. The `/jlens` viewer does **no** linear algebra — all quantities are precomputed offline and shipped as static JSON.

Auditability. Every quantitative claim resolves through the audit ledger (Appendix D, `jlens/paper/build_ledger.py`) to a source bundle and JSON path; every method claim resolves through `jlens/paper/method_map.md` to a `file:line`.

Appendix A — δ-broadcast derivation

A masked-mean-pooled encoder outputs $p = \frac{1}{T}\sum_t h_{L,t}$, the mean over T positions of the final layer. The full sensitivity of p to the layer-ℓ residuals is a rank-4 object $\partial p_i/\partial h_{\ell,t,j}$ of size $d \times (T \times d)$ — intractable to form and to average across variable-length prompts. We collapse it with a **shared perturbation**: perturb *every* source position by the same $\delta \in \mathbb{R}^d$, $h_{\ell,t} \mapsto h_{\ell,t} + \delta$, and differentiate the pooled output,

$$J_\ell \;=\; \frac{\partial p}{\partial \delta}\Big|_{\delta=0} \;=\; \frac{1}{T}\sum_t \frac{\partial p}{\partial h_{\ell,t}} \;\in\; \mathbb{R}^{d\times d},$$

one $d \times d$ matrix per layer, independent of T. Each column c is estimated by **central finite differences**, $J_{\cdot c} \approx \frac{p(+\varepsilon e_c) - p(-\varepsilon e_c)}{2\varepsilon}$ (with $\pm\varepsilon e_c$ broadcast to all positions), whose error is $O(\varepsilon^2)$ by Taylor expansion; columns are computed in batches (pure gemm) via one `forward_from(ℓ)` per batch. On a causal decoder (§5.5) the identical construction reads the **last** position $p = h_{L,\text{last}}$ (the next-token driver) rather than the mean, giving the causal analog. The embedding-space projection (§4.3) removes the component along the unit output $e = p/\|p\|$, since a normalized embedding is invariant to radial scaling: $J_{\text{emb}} = \frac{1}{\|p\|}(I - ee^\top)J$.

Appendix B — Per-layer tables

Structural signatures and internal identity accuracy by residual depth, straight from the ledger.

Authors (MiniLM, 6 layers) — internal identity accuracy by layer: [5.2, 7.2, 8.8, 8.0, 7.2, 8.8]% (chance 1.8%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	22.1	73.5	0.50	-0.004
1	31.4	88.8	0.51	+0.004
2	31.2	93.0	0.53	+0.048
3	37.2	104.6	0.50	+0.101

layer	stable rank	eff dim	verbaliz.	autocorr
4	54.5	127.6	0.49	+0.116
5	106.9	205.2	0.61	+0.088

Coders (JinaBERT, 12 layers, 15 developers) — internal identity accuracy by layer: [45.2, 44.8, 48.0, 50.8, 40.0, 48.8, 48.0, 40.4, 42.0, 42.0, 42.4, 43.2]% (chance 6.7%).

layer	stable rank	eff dim	verbaliz.	autocorr
0	30.6	79.6	0.19	-0.033
1	36.7	95.2	0.20	-0.030
2	38.6	98.9	0.21	-0.001
3	35.6	100.7	0.21	+0.023
4	33.4	102.4	0.21	+0.032
5	20.3	91.2	0.21	+0.039
6	37.9	113.8	0.19	+0.062
7	54.3	126.4	0.20	+0.071
8	59.5	132.0	0.20	+0.086
9	61.7	139.0	0.22	+0.085
10	73.3	151.3	0.23	+0.088
11	79.0	185.0	0.22	+0.075

Decoder (GPT-2, 12 layers) — next-token accuracy by residual depth: [0.0, 2.6, 2.6, 3.2, 3.2, 4.7, 5.1, 7.2, 10.9, 13.0, 17.0, 21.3, 22.6]%.

layer	stable rank	eff dim	verbaliz.	autocorr
0	2.6	5.8	0.89	+0.007
1	5.0	16.2	1.81	+0.050
2	6.3	25.8	2.44	+0.031
3	6.4	30.3	2.08	+0.047
4	6.6	32.1	2.15	+0.044
5	6.7	33.5	1.94	+0.052
6	6.5	31.1	1.90	+0.058
7	6.5	29.9	1.57	+0.099
8	6.7	32.5	1.26	+0.123
9	5.2	23.7	1.39	+0.145
10	3.8	13.8	1.10	+0.161
11	2.0	4.1	0.89	+0.110

Appendix C — Axis definitions, rosters, and OOD probe sets

Prose style axes (`author_axes`): `male↔female` (gender difference-of-means); and the top-2 one-vs-rest classes of the `educated` and `raised` fields — `educated=England↔rest`, `educated=US↔rest`, `raised=England↔rest`, `raised=US↔rest`. **Code style axes** (`dataset_axes`): `Systems↔rest`, `Scripting↔rest` (the `paradigm` trait). Each axis is the L2-normalized difference of class-mean reference embeddings — a unit Fisher direction, used for both readout and steering.

Rosters. 55 public-domain prose authors (`corpus/authors.toml`, labelled with gender / birth / raised / educated / college) and 15 open-source developers (`corpus/coders.toml`, git-attributed and **name-scrubbed**, labelled with a `paradigm` = Systems/Scripting trait).

OOD probes for the fingerprint detector (§5.4). Prose (bar 0.30): four held-out author samples plus `source code`, `modern chat`, `biology abstract`, `legalese`. Code (bar 0.51): four held-out coder samples plus `Victorian prose`, `modern chat`, `news headline`, `recipe`.

Appendix D — Audit ledger

Every number above is generated by `jlens/paper/build_ledger.py` into `jlens/paper/ledger.json` (value → source asset + JSON path) and cross-checked by the figure pipeline. Method claims resolve via `jlens/paper/method_map.md`.